

Markov □ Kalman Filter Localization

- Markov localization

- localization starting from any unknown position
- recovers from ambiguous situation.
- However, to update the probability of all positions within the whole state space at any time requires a discrete representation of the space (grid). The required memory and calculation power can thus become very important if a fine grid is used.

- Kalman filter localization

- tracks the robot and is inherently very precise and efficient.
- However, if the uncertainty of the robot becomes too large (e.g. collision with an object) the Kalman filter will fail and the position is definitively lost.